

Sockeye / Blurasaga AI Services Master Documentation

Comprehensive Architecture, Logic, API Specifications & Integration Guide for OCR, Sentiment API, and LLM Services

Table of Contents

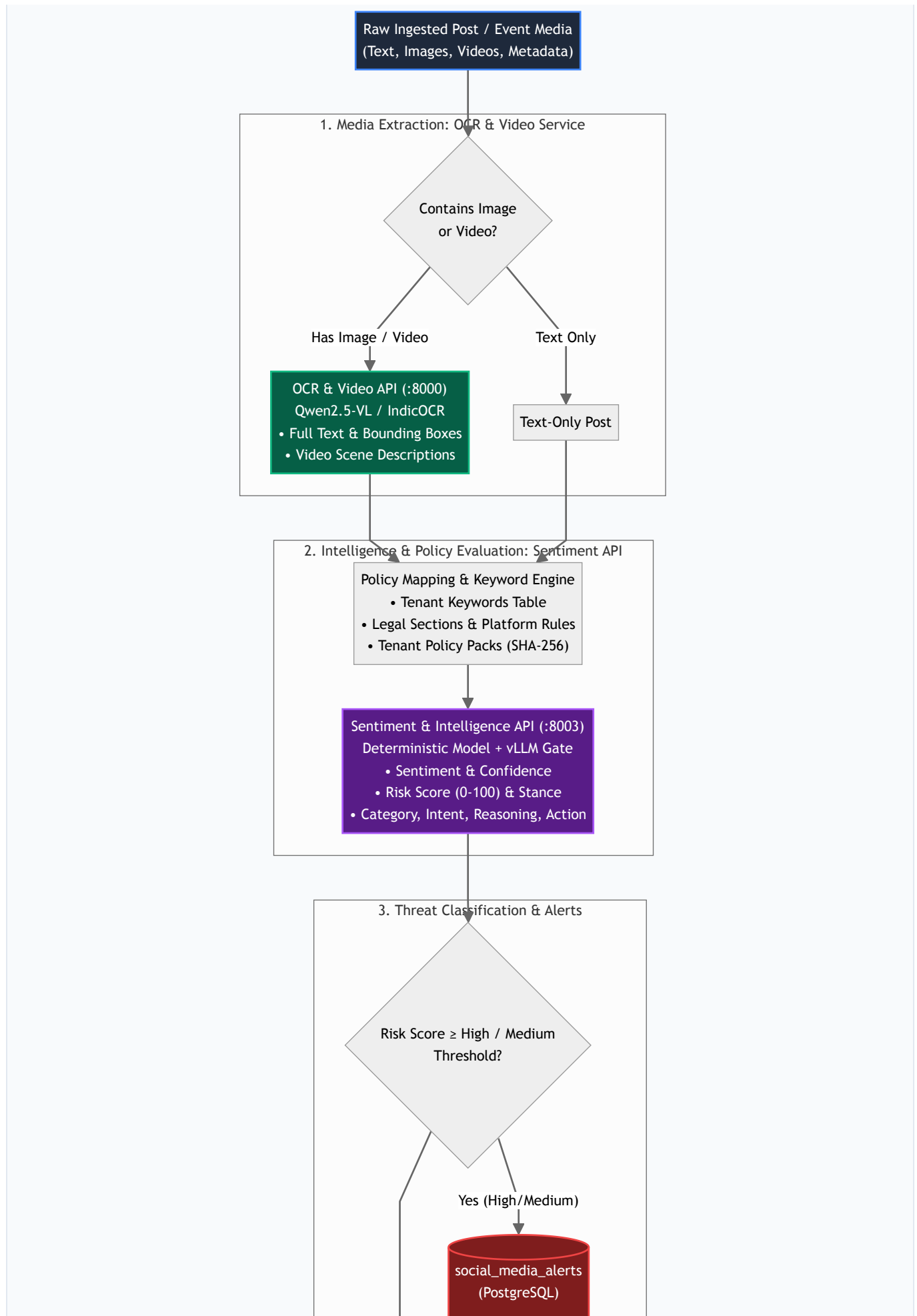
- 1. [Executive Overview & Integrated Architecture](#)
- 2. [Service 1: OCR & Video Analysis Service](#)
 - [2.1 Purpose & Use Cases](#)
 - [2.2 Architecture & Underlying Engine](#)
 - [2.3 Client Implementation & Execution Logic](#)
 - [2.4 API Endpoint Specifications & Payloads](#)
 - [2.5 Error Handling, Retries & Fallbacks](#)
- 3. [Service 2: Custom Sentiment & Intelligence API](#)
 - [3.1 Purpose & Capabilities](#)
 - [3.2 Architecture & Multimodal Pipeline](#)
 - [3.3 Concurrency Lanes & Tenant-Fair Admission Gate](#)
 - [3.4 Policy Pack Sync & Dynamic Keyword Injection](#)
 - [3.5 Execution Logic & Alert Generation Flow](#)
 - [3.6 API Endpoint Specifications & Response Schema](#)
- 4. [Service 3: Large Language Model \(LLM\) Service & Summarization](#)
 - [4.1 Purpose & Core Roles](#)
 - [4.2 Architecture & Infrastructure \(vLLM / Ollama\)](#)
 - [4.3 Event Intelligence Summarizer \(Single-Pass vs Map-Reduce\)](#)
 - [4.4 Auxiliary LLM Services: Event Terms & Hashtag Generator](#)
 - [4.5 UTF-8 Sanitization & Post-Processing Logic](#)
 - [4.6 API Endpoint Specifications & Prompt Structures](#)
- 5. [Comparative Integration Matrix](#)
- 6. [Complete End-to-End Processing Workflow](#)
- 7. [Environment Variables & Configuration Reference](#)
- 8. [Operational Runbook & Troubleshooting](#)

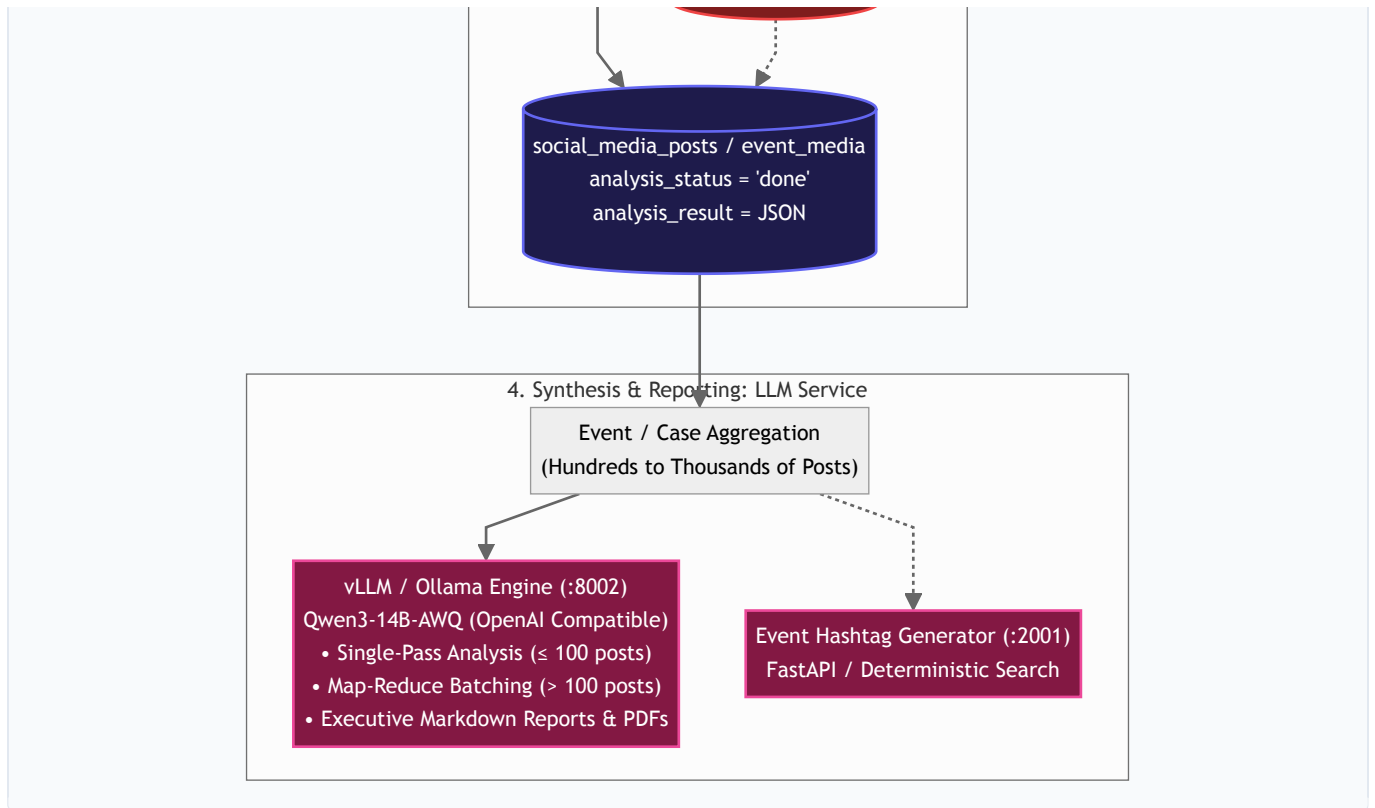
1. Executive Overview & Integrated Architecture

The **Blurasaga / Sockeye** intelligence platform processes large volumes of unstructured multi-platform social media data (X/Twitter, Facebook, Instagram, YouTube, Telegram, Reddit, RSS) across multiple isolated tenant databases.

To transform raw posts into actionable intelligence and real-time security alerts, the system integrates three specialized AI services:







2. Service 1: OCR & Video Analysis Service

2.1 Purpose & Use Cases

A massive portion of intelligence signals on social media exists inside images and videos (e.g., protest banners, newspaper cutouts, meme text, official notices, on-screen video text, incident footage). Standard text scrapers miss this content completely.

The **OCR & Video Analysis Service** solves this by:

- High-Fidelity Text Extraction (Images):** Reads English, Hindi, Odia, and other Indic scripts directly from uploaded or scraped images.
- Video Scene Understanding (Videos):** Samples video frames across video clips, extracting visual scene descriptions, contextual summaries, and on-screen textual overlays.
- Preserving Raw Post Integrity:** Stores extracted text separately in `image_analysis` rather than appending directly into the raw post text, maintaining raw data fidelity for legal audits.

2.2 Architecture & Underlying Engine

- Service Endpoint:** `http://101.53.140.97:8000` (Configured via `OCR_SERVICE_URL`).
- Primary Vision-Language Model:** Qwen/Qwen2.5-VL-7B-Instruct served via vLLM on port `8001` (`http://101.53.140.97:8001`).
- Fallback Engine:** Local `IndicOCR` multi-threaded worker pool for OCR extraction when GPU VLM is busy or degraded.
- Hardware Acceleration:** Shared NVIDIA L40S 46 GB GPU (~27.8 GB dedicated to Qwen2.5-VL).

2.3 Client Implementation & Execution Logic

Implemented in `backend/src/services/media_post_analysis/extract0cr.js` and consumed by `analyzeMediaPost.js`.

1. Image Download with Anti-403 Referer Spoofing

CDN providers (Instagram, Facebook/Meta CDN, X/Twitter `twimg.com`, Telegram) block automated image scrapers with HTTP `403 Forbidden` if headers are missing. The client dynamically derives the proper `Referer` and `User-Agent`:

```
function refererFor(url) {
  try {
    const { hostname } = new URL(url);
    if (/instagram\.com|cdninstagram\.com/i.test(hostname)) return 'https://www.instagram.com/';
    if (/fbcdn\.net|facebook\.com/i.test(hostname)) return 'https://www.facebook.com/';
    if (/twimg\.com|twitter\.com|x\.com/i.test(hostname)) return 'https://twitter.com/';
    if (/telegram\.org|t\.me/i.test(hostname)) return 'https://web.telegram.org/';
    return undefined;
  } catch (_) {
    return undefined;
  }
}
```

2. Base64 Image Conversion

Images up to **25 MB** are downloaded into memory buffers and converted to Base64 strings:

```
const resp = await axios.get(imageUrl, {
  responseType: 'arraybuffer',
  timeout: 30000,
  headers: { 'User-Agent': 'Mozilla/5.0 ...', ...(referer ? { Referer: referer } : {}) },
  maxContentLength: 25 * 1024 * 1024,
});
const base64Image = Buffer.from(resp.data).toString('base64');
```

3. Video URL Extraction Strategy

Direct binary upload of large video files would exhaust network sockets. Therefore, for videos:

- The backend identifies direct video stream URLs (`.mp4` , `.mov` , `.webm` , `.m3u8`).
- The backend sends `{ video_url, tenant_key }` directly to the OCR service.
- The OCR service downloads the video server-side using SSRF-guarded fetchers, extracts sampled keyframes, and runs Qwen2.5-VL inference.

2.4 API Endpoint Specifications & Payloads

Endpoint 1: Health & Liveness

- **Method:** `GET /health`
- **Response:**

```
{
  "status": "healthy",
  "active_engine": "qwen2.5-vl",
  "indic_fallback_pool": {
    "pool_size": 0,
    "workers_available": 0
  }
}
```

Endpoint 2: Extract Text / Video Descriptions

- **Method:** `POST /extract`
- **Headers:** `Content-Type: application/json`
- **Request Body Options:**

Option A: Image (Base64)

```
{
  "image_base64": "/9j/4AAQSkZJRgABAQAAQABAA...",
  "tenant_key": "blurasaga_user_police_1",
  "min_confidence": 0.6
}
```

Option B: Video (Direct URL)

```
{
  "video_url": "https://video.twimg.com/ext_tw_video/12345/pu/vid/720x1280/xyz.mp4",
  "tenant_key": "blurasaga_user_police_1"
}
```

Response Structure:

```
{
  "success": true,
  "data": {
    "full_text": "PROTEST NOTICE: Gathering at Master Canteen Square on 10th Oct at 10:00 AM.",
    "bounding_boxes": [
      {
        "box": [120, 45, 600, 180],
        "text": "PROTEST NOTICE",
        "confidence": 0.98
      }
    ],
    "language": "en",
    "confidence": 0.96,
    "description": "A yellow protest poster calling for an assembly at Master Canteen.",
    "summary": "Public protest announcement poster.",
    "visible_text": ["PROTEST NOTICE", "Master Canteen", "10th Oct"]
  },
  "error": null
}
```

2.5 Error Handling, Retries & Fallbacks

- **Retry Mechanism:** Exponential backoff (`1s` , `2s` , `4s` ... up to `8s`).
- **Max Attempts:** Configurable via `OCR_MAX_ATTEMPTS` (Default: `3` for images, `2` for videos).

- **Non-retryable Errors:** HTTP `403 Forbidden` / `404 Not Found` on image download fail fast immediately to avoid stalling the processing queue.
- **Failures Saved Gracefully:**

```
{
  "error": "Download failed: Request failed with status code 404",
  "failed": true,
  "attempts": 1,
  "media_type": "image"
}
```

3. Service 2: Custom Sentiment & Intelligence API

3.1 Purpose & Capabilities

Social media text often contains sarcasm, code-mixed Indic languages (Hinglish/Odia-English), subtle threats, or administrative grievances that simple keyword filters cannot accurately categorize.

The **Sentiment & Intelligence API** provides deep text understanding:

1. **Deterministic Sentiment Analysis:** High-speed, reproducible 3-class sentiment (`positive` , `neutral` , `negative`) with confidence scores.
2. **Stance Detection:** Determines whether the author **Supports**, **Opposes**, is **Neutral**, or **Unclear** towards a specified government body, leader, or event topic.
3. **Threat & Risk Scoring (0-100):** Quantifies actionable risk based on violence incitement, communal disharmony, defamation, public order risks, or official complaints.
4. **Intent & Policy Categorization:** Maps posts to predefined legal and administrative violation categories (e.g., *Defamation*, *Hate Speech*, *Public Order*, *Administrative Grievance*).
5. **LLM Chain-of-Thought Reasoning & Action:** Provides human-readable reasoning and recommended police/administrative action.

3.2 Architecture & Multimodal Pipeline

- **Service Endpoint:** `http://127.0.0.1:8003/analyze/intelligence` (or `http://101.53.140.97:8003`).
- **Internal Pipeline:**
 - **Stage 1:** Language identification, translation/transliteration to English if needed.
 - **Stage 2:** Deterministic sentiment engine (Cardiff NLP roBERTa).
 - **Stage 3:** LLM inference gate for Category, Risk, Stance, and Reasoning.

3.3 Concurrency Lanes & Tenant-Fair Admission Gate

Implemented in `backend/src/modules/intelligence/intelligence.client.service.js` .

To prevent background batch ingestion from starving interactive user UI requests (e.g., manual single-post analysis or live investigation), the client implements **Two Independent Lanes**:

Metric / Setting	Bulk Lane (<code>bulk</code>)	Interactive Lane (<code>interactive</code>)
Primary Use Case	Scheduled background pollers, full dataset re-analysis	User clicking "Analyze Now", single post triage
Target Concurrency	Configurable via <code>INTELLIGENCE_BULK_CONCURRENCY</code> (Default: <code>1</code>)	Configurable via <code>INTELLIGENCE_INTERACTIVE_CONCURRENCY</code> (Default: <code>1</code>)
Request Timeout	<code>180,000 ms</code> (3 minutes)	<code>100,000 ms</code> (~1.6 minutes)
Ready Queue Capacity	<code>200</code> items	<code>50</code> items
Max Waiters (FIFO)	<code>5,000</code> items	<code>500</code> items
Drop Behavior	Zero data loss; parks in FIFO waiter pool until worker slot opens	Dedicated pool ensures instant response for UI users

Tenant-Fair Scheduling (`tenant_key` & `tenant_name`):

- `tenant_key` (`dbName`): Ensures multiple tenants sharing the same backend do not starve each other in the admission queue.
- `tenant_name` : Passes the sanitized admin/tenant display name (e.g., *"Odisha Police Crime Branch"*) to ground stance detection.

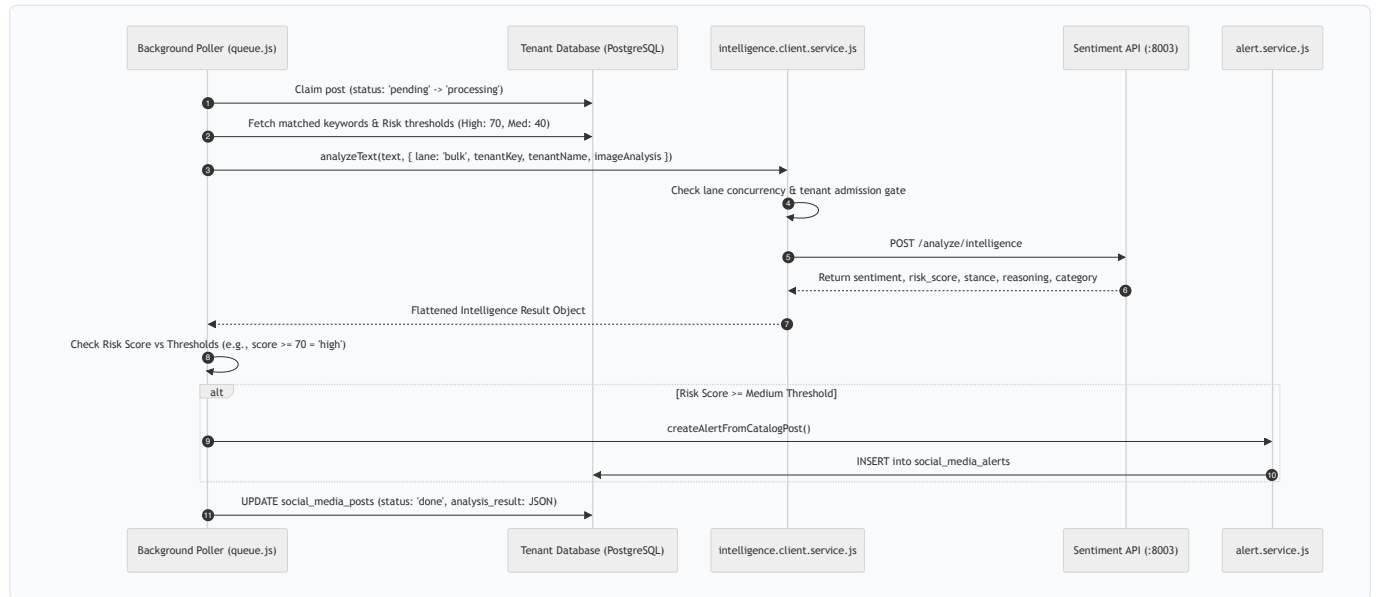
3.4 Policy Pack Sync & Dynamic Keyword Injection

The client extracts active categories from the database via `mappingService` and constructs a structured **Policy Pack** sent to the Sentiment API:

```
function buildPolicyPack() {
  const categories = mappingService.mappingData.category_mappings.map(m => ({
    id: m.category_id,
    definition: m.definition.slice(0, 500),
    severity: m.severity_level,
    keywords: m.keywords.slice(0, 40)
  }));
  // Uses SHA-256 fingerprint caching to avoid redundant JSON overhead
  const fingerprint = `sha256:${crypto.createHash('sha256').update(JSON.stringify(categories)).digest('hex')}`;
  return { name: 'sockeye-policy-mapping', fingerprint, categories };
}
```

3.5 Execution Logic & Alert Generation Flow

Implemented in [analyzePost.js](#) and [analyzeMediaPost.js](#).



3.6 API Endpoint Specifications & Response Schema

Endpoint: Intelligence Analysis

- **Method:** POST /analyze/intelligence
- **Headers:**
 - Content-Type: application/json
 - x-api-key: <GATEWAY_API_KEY>

Request Payload:

```

{
  "texts": ["Heavy protest planned outside SP office tomorrow regarding electricity tariff hike."],
  "tenant_name": "State Police Intelligence Wing",
  "tenant_key": "blurasaga_sp_office_1",
  "policy_pack": {
    "name": "sockeye-policy-mapping",
    "fingerprint": "sha256:8f4c28...",
    "unknown_label": "Normal",
    "categories": [
      {
        "id": "Public Unrest",
        "definition": "Calls for mass agitations, road blockades, or vandalism",
        "severity": "High",
        "keywords": ["protest", "strike", "blockade", "gherao"]
      }
    ]
  },
  "image_analysis": {
    "full_text": "RALLY TOMORROW 10AM",
    "language": "en"
  },
  "intent_mode": "free",
  "timeout_s": 120
}
  
```

Response Structure:

```
{
  "results": [
    {
      "sentiment": "negative",
      "confidence": 0.94,
      "language": "en",
      "english_text": null,
      "was_translated": false,
      "was_transliterated": false,
      "intelligence": {
        "risk_score": 75,
        "stance": "oppose",
        "stance_confidence": 0.89,
        "category": "Public Unrest",
        "intent": "Mobilizing public agitation against administration",
        "reasoning": "Post explicitly organizes a protest outside a key government installation (SP office) targeting public officials.",
        "summary": "Protest call against electricity tariff hike outside SP office.",
        "recommended_action": "Deploy preventative security personnel and monitor local organizers.",
        "signals": ["protest_organizing", "government_target"],
        "evidence_confidence": "high",
        "source": "provider",
        "model": "qwen3-14b-intelligence",
        "latency_ms": 1420
      }
    }
  ]
}
```

4. Service 3: Large Language Model (LLM) Service & Summarization

4.1 Purpose & Core Roles

While the Sentiment API operates at a single-post granularity, the **LLM Service** operates at the **Macro / Aggregate Level**. It synthesizes hundreds or thousands of posts collected during an event or incident into high-level analytical reports:

- 1. **Event Executive Summaries:** Generates comprehensive situation reports covering chronological timeline, grievance highlights, sentiment breakdown, key influencers, and threat levels.
- 2. **Automated PDF Situation Reports:** Formats narrative intelligence with citations of specific post IDs, author handles, and engagement numbers for high-ranking officials.
- 3. **Keyword & Hashtag Generator:** Generates relevant monitoring hashtags and boolean search terms from event title, location, and description.

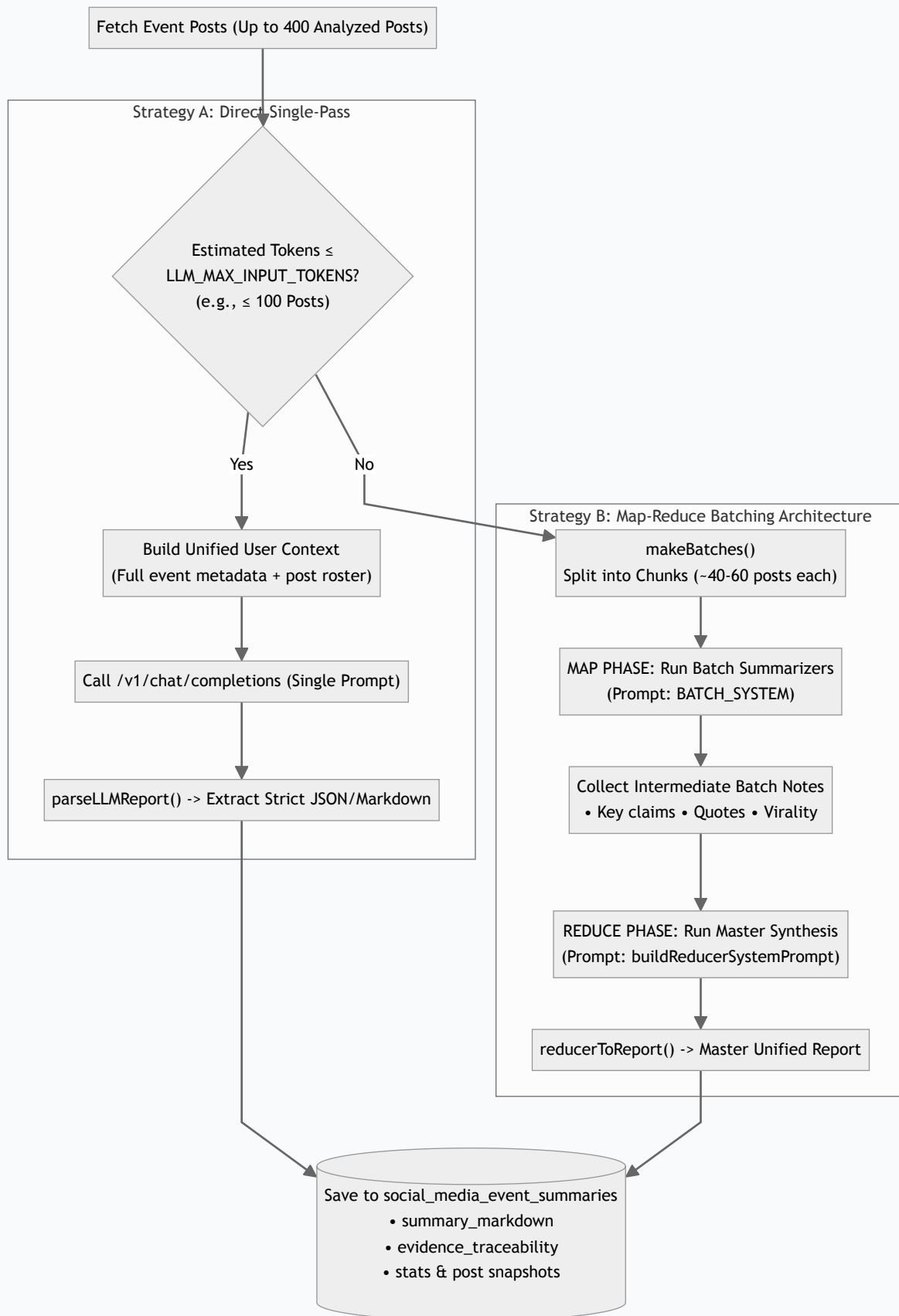
4.2 Architecture & Infrastructure (vLLM / Ollama)

- **Primary Host:** `http://101.53.140.97:8002` (Configured via `LLM_BASE_URL` with `/v1/chat/completions`).
- **Model:** `Qwen/Qwen3-14B-AWQ` (OpenAI-compatible protocol).
- **Alternative / Local Engine:** Ollama running locally on port `11434` (`OLLAMA_BASE_URL=http://127.0.0.1:11434`).
- **Memory Footprint:** Fits within ~13.1 GB GPU VRAM on the NVIDIA L40S, with up to `4` concurrent sequences.

4.3 Event Intelligence Summarizer (Single-Pass vs Map-Reduce)

Implemented in `backend/src/services/SummaryLLM/eventSummary.service.js` and `eventSummary.prompt.js`.

Depending on the number of posts associated with an event, the system automatically selects one of two processing strategies:

**Token Safety & Budgeting:**

- `LLM_CONTEXT_WINDOW` : Defaults to 16,384 tokens.
- `LLM_MAX_INPUT_TOKENS` : Defaults to 12,000 tokens.
- `LLM_SUMMARY_MAX_TOKENS` : Output ceiling bounded between 1,500 and 4,000 tokens (Default: 3,000).

4.4 Auxiliary LLM Services: Event Terms & Hashtag Generator

Implemented in `backend/src/services/eventHashtag/eventHashtag.client.js`.

When an officer creates an event in the UI (e.g., *"Rath Yatra 2026 at Puri"*), the hashtag generator searches web and social corpus to recommend high-precision search keywords:

- **Upstream Service:** `http://127.0.0.1:2001` (Managed by PM2 as `event-hashtag-generator`).
- **Endpoint:** `POST /event-terms`
- **Payload:**

```
{
  "event": "Farmers Protest Rally",
  "location": "Bhubaneswar, Odisha",
  "description": "Kisan Mahapanchayat demanding minimum support price revision.",
  "include_social": true,
  "max_hashtags": 10,
  "max_keywords": 15
}
```

- **Response:**

```
{
  "hashtags": ["#KisanProtest", "#OdishaFarmers", "#BhubaneswarRally", "#MSPNow"],
  "keywords": ["kisan mahapanchayat", "farmer union", "PM Fasal Bima", "krushak morcha"]
}
```

4.5 UTF-8 Sanitization & Post-Processing Logic

LLM tokenizers and PostgreSQL JSONB columns crash on lone surrogate pairs, non-printable control characters, or unescaped null bytes. The service enforces strict pre- and post-sanitization:

```
const cleanSafeUtf8 = (val, maxlen = 0) => {
  if (val === null || val === undefined) return '';
  let s = String(val);
  // Replaces unpaired surrogates with well-formed unicode
  if (typeof s.toWellFormed === 'function') {
    s = s.toWellFormed();
  } else {
    s = s.replace(/[\uD800-\uDBFF](?![\uDC00-\uDFFF])|(?![\uD800-\uDBFF][\uDC00-\uDFFF])/g, '');
  }
  // Strips null bytes and non-printable control characters
  s = s.replace(/[\x00-\x08\x0B\x0C\x0E-\x1F\x7F]/g, '');
  s = s.replace(/[\r\n\t]/g, ' ').replace(/s+/g, ' ').trim();
  return maxlen > 0 ? s.slice(0, maxlen) : s;
};
```

4.6 API Endpoint Specifications & Prompt Structures

Chat Completions Request Format (OpenAI Standard)

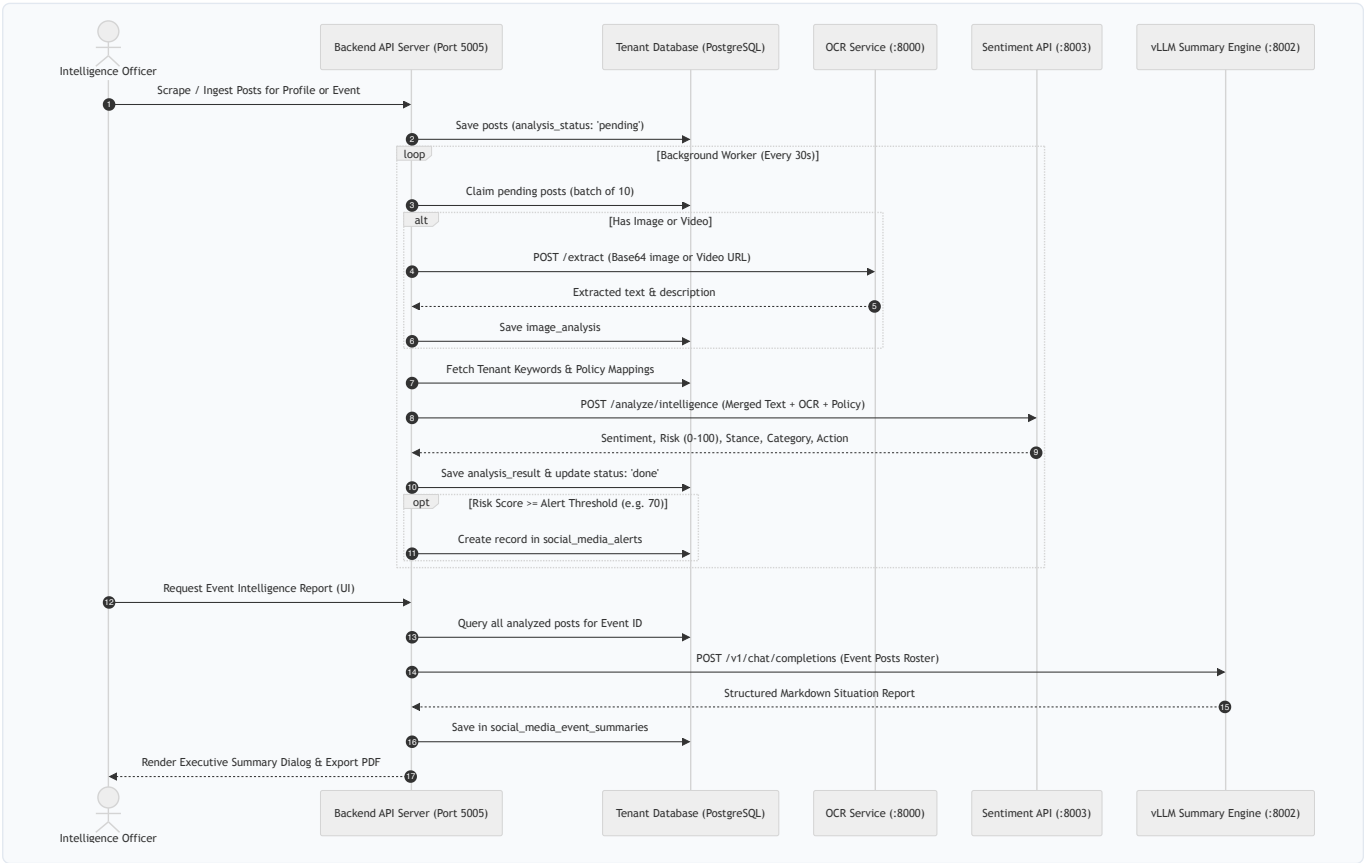
- **Endpoint:** `POST {LLM_BASE_URL}/chat/completions`
- **Payload:**

```
{
  "model": "qwen3-14b",
  "temperature": 0.2,
  "max_tokens": 3000,
  "messages": [
    {
      "role": "system",
      "content": "You are a senior Law Enforcement & Intelligence Analyst. Produce an executive situation report adhering strictly to verified evidence..."
    },
    {
      "role": "user",
      "content": "EVENT CONTEXT: ... \nPOST ROSTER: \n[ID: 10492] @handle: Protest begins at 10 AM... (Likes: 450, Reposts: 120)"
    }
  ]
}
```

5. Comparative Integration Matrix

Dimension	OCR & Video Service	Sentiment & Intelligence API	LLM Summarization Service
Primary Base URL	http://101.53.140.97:8000	http://127.0.0.1:8003 / http://101.53.140.97:8003	http://101.53.140.97:8002/v1
Primary Environment Variable	OCR_SERVICE_URL	CUSTOM_SENTIMENT_URL / INTELLIGENCE_SERVICE_URL	LLM_BASE_URL
Underlying Models	Qwen2.5-VL-7B-Instruct + IndicOCR	Cardiff roBERTa + vLLM Qwen Gate	Qwen3-14B-AWQ (vLLM / Ollama)
Calling Module	media_post_analysis/extractocr.js	modules/intelligence/intelligence.client.service.js	services/SummaryLLM/eventSummary.service.js
Primary Input	Image Base64 / Direct Video URL	Cleaned Post Text + OCR Text + Policy Pack + Keywords	Hundreds of aggregated posts + Event metadata
Primary Output	Extracted Text, Bounding Boxes, Video Summary	Sentiment, Risk Score (0-100), Stance, Category, Reasoning	Executive Markdown Narrative, Timeline, PDF Report
Target Database Columns	social_media_posts.image_analysis	social_media_posts.analysis_result	social_media_event_summaries.summary_markdown
Retry Strategy	Exponential backoff (1s, 2s, 4s...) up to 3 tries	Exponential backoff up to 5 tries; FIFO waiter pool on 429	Map-Reduce sub-batch retry on timeout/failure
Failure Behavior	Marks image_analysis.failed = true , pipeline continues	Re-queues up to MAX_ATTEMPTS ; marks failed if exhausted	Fallback to deterministic statistical summary

6. Complete End-to-End Processing Workflow



7. Environment Variables & Configuration Reference

Add or verify these environment variables in `backend/.env` :

```
# =====
# AI & MACHINE LEARNING SERVICES CONFIGURATION
# =====

# 1. OCR & Video Analysis Service (saga GPU box)
OCR_SERVICE_URL=http://101.53.140.97:8000
OCR_MAX_ATTEMPTS=3
OCR_VIDEO_MAX_ATTEMPTS=2
OCR_VIDEO_TIMEOUT_MS=600000

# 2. Custom Sentiment & Intelligence API
CUSTOM_SENTIMENT_URL=http://127.0.0.1:8003
INTELLIGENCE_SERVICE_URL=http://127.0.0.1:8003
SENTIMENT_MAX_ATTEMPTS=5
INTELLIGENCE_BULK_CONCURRENCY=1
INTELLIGENCE_INTERACTIVE_CONCURRENCY=1
INTELLIGENCE_TIMEOUT_MS=180000
INTELLIGENCE_INTENT_MODE=free

# 3. LLM Service (vLLM / Ollama for Event Summaries & Reports)
LLM_BASE_URL=http://101.53.140.97:8002/v1
LLM_API_KEY=ollama
LLM_MODEL=qwen3-14b
LLM_TIMEOUT_MS=180000
LLM_SUMMARY_MAX_TOKENS=3000
LLM_CONTEXT_WINDOW=16384
LLM_MAX_INPUT_TOKENS=12000
LLM_BATCH_CONCURRENCY=1

# 4. Event Keyword & Hashtag Generator (PM2 Local Process)
EVENT_HASHTAG_GENERATOR_URL=http://127.0.0.1:2001
EVENT_HASHTAG_GENERATOR_TIMEOUT_MS=120000
```

8. Operational Runbook & Troubleshooting

Scenario 1: OCR Service is Timing Out or Failing

1. Check OCR service health:

```
curl -s http://101.53.140.97:8000/health | jq
```

2. Check underlying Qwen2.5-VL container on port 8001:

```
curl -s http://101.53.140.97:8001/health
```

3. If images fail with HTTP 403, verify if the domain is missing in `refererFor()` in `extract0cr.js`.

Scenario 2: Sentiment API Returning 429 Backpressure

- The client automatically catches HTTP 429 responses and pauses subsequent requests according to the `Retry-After` header.
- Posts remain in `pending` status in PostgreSQL and are safely picked up during the next polling tick without data loss.

Scenario 3: LLM Context Length Exceeded on Large Events

- The event summarizer uses `makeBatches()` to slice posts into chunks of 40–60 items.
- If individual batch prompts exceed token limits, reduce `LLM_MAX_ANALYSED_POSTS` in `.env` (default is `400`).

Document Information

- **Applies to:** Sockeye / Blurasaga Intelligence Platform
- **Primary Source Code References:**
 - OCR: `backend/src/services/media_post_analysis/extract0cr.js`
 - Sentiment: `backend/src/modules/intelligence/intelligence.client.service.js`
 - LLM: `backend/src/services/SummaryLLM/eventSummary.service.js`
 - Hashtags: `backend/src/services/eventHashtag/eventHashtag.client.js`